

Introduction

The Olympic Games are shaped by **social, cultural, historical, and institutional forces** that influence who participates and at times wins medals. Access to sport varies across gender, class, race, and nationality, and the gradual inclusion of previously excluded groups such as women has changed participation and medal distributions over time. Cultural values and national priorities shape investment in particular sports, while historical events such as wars and **geopolitical tensions** have influenced Olympic participation and outcomes. The International Olympic Committee (IOC) and National Olympic Committees (NOCs) also shape the Games through eligibility rules, sport inclusion, and competition standards.

This project analyzes a dataset of Olympic athlete results from **Athens 1896 to Rio 2016** containing over **270,000** athlete event records. These records were scraped from Sports Reference (an abandoned community-driven collection effort). Each observation represents one athlete competing in one Olympic event and includes variables such as age, sex, height, weight, country, sport, year, host city, and medal outcome. **Our research asks whether athlete characteristics can help predict the likelihood of winning an Olympic medal despite the extreme imbalance between medalists and non-medalists.**

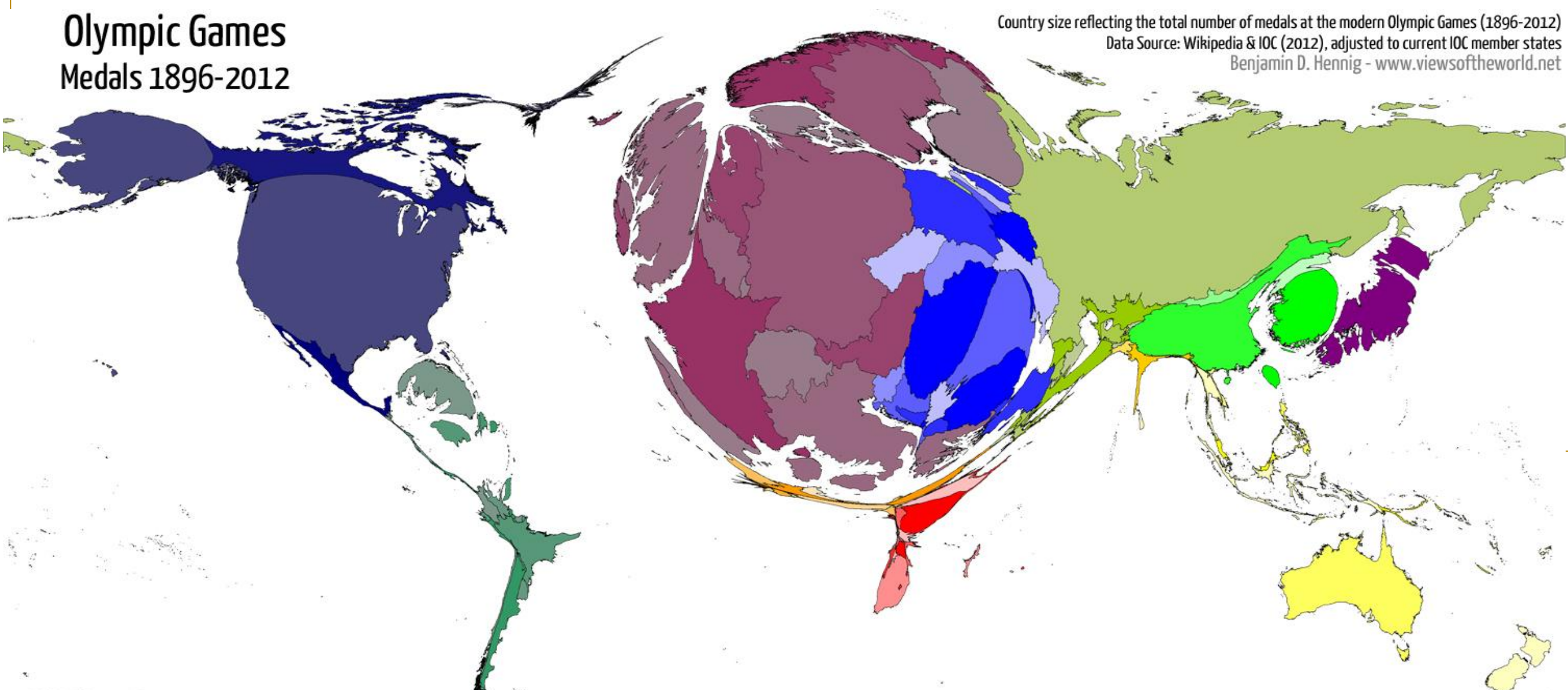


Fig 1: Country size reflecting the total number of medals at the modern Olympic Games (1896-2012), Source: Benjamin D. Hennig - www.viewsoftheworld.net

Methods

- Data Cleaning:** We employed some basic modifications to the data to make it easier for our model to fit, including:
- Made the ‘Medal’ variable binary (1 - Medal or 0 - No Medal) to make the classification task easier for our models.
 - Split up the large ‘Athletics’ sport category, which has all of the track and field events, into sub sports based on the *Event* type (Sprints, Distance, Throws, Jumps, etc.)
 - Removed predictors that were not useful (*ID*, *Name*) or that provide redundant/overlapping information to our models
 - For example, *Team* and *NOC* (National Olympic Committee) both point to the athlete’s home nation, but we found *NOC* to be more reliable because some nations have changed names through history.
 - Note: we also removed *Event* as we felt it had too many unique classes to be one-hot encoded without feature explosion.
- Models:** We ran large meta-grid search with 5-fold cross-validation, in order to compare the performance of Logistic Regression models against Decision Tree models while tuning both model hyperparameters separately.
- For each model, we use a basic preprocessing pipeline to standardize numeric variables, impute missing values with the median, and one-hot encode categorical variables.
 - We established F1 score as our main evaluation metric as we felt that it would give us a better overall picture of model performance given the extreme class imbalance in the data.
 - Mean CV, Train, and Test F1 are all recorded at every iteration for verbosity.

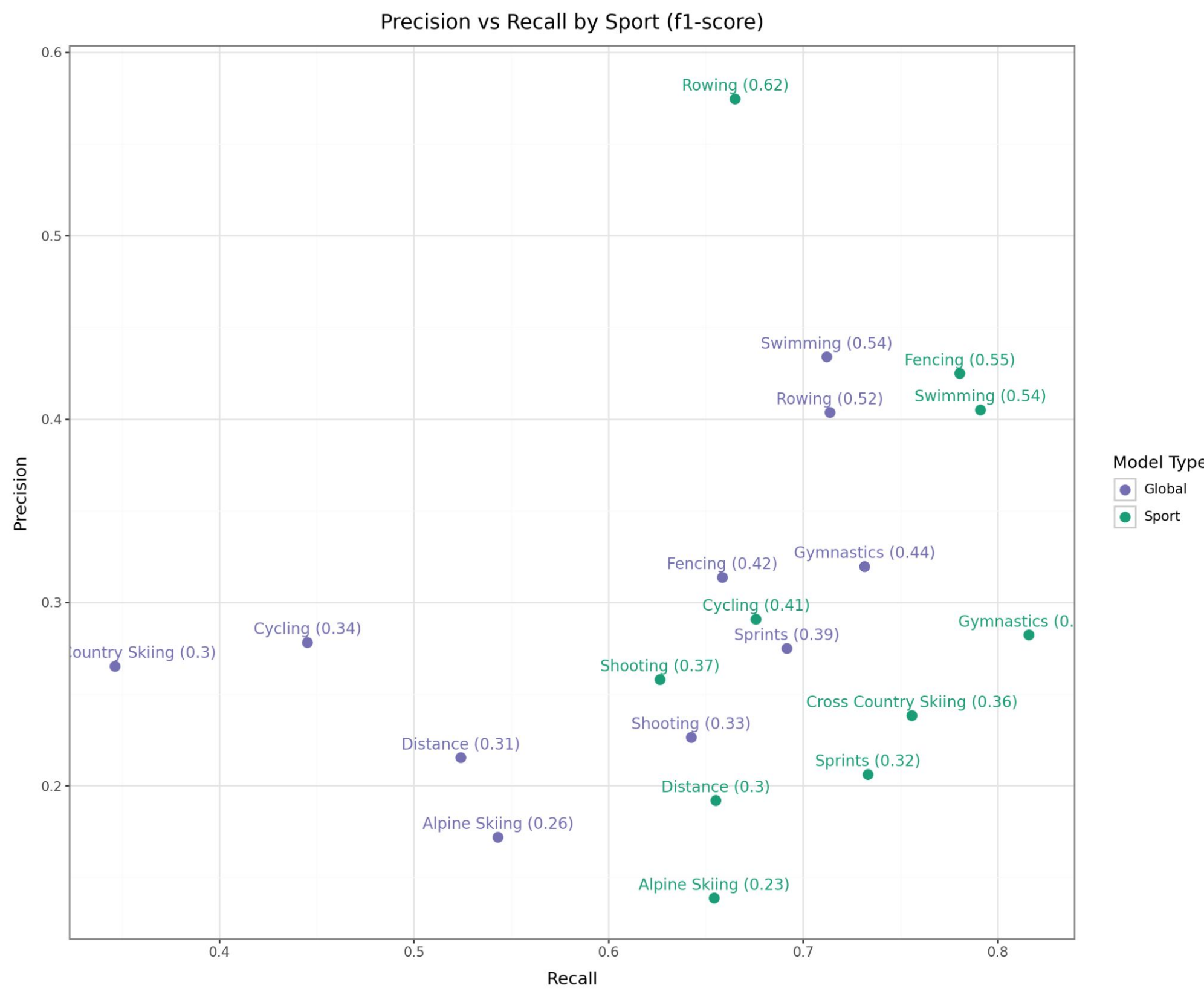


Fig 2: Comparison of the performance of our global model vs. our sport-specific models for the top 10 most frequent sports in the dataset. Both use the same decision tree model setup as discussed below.

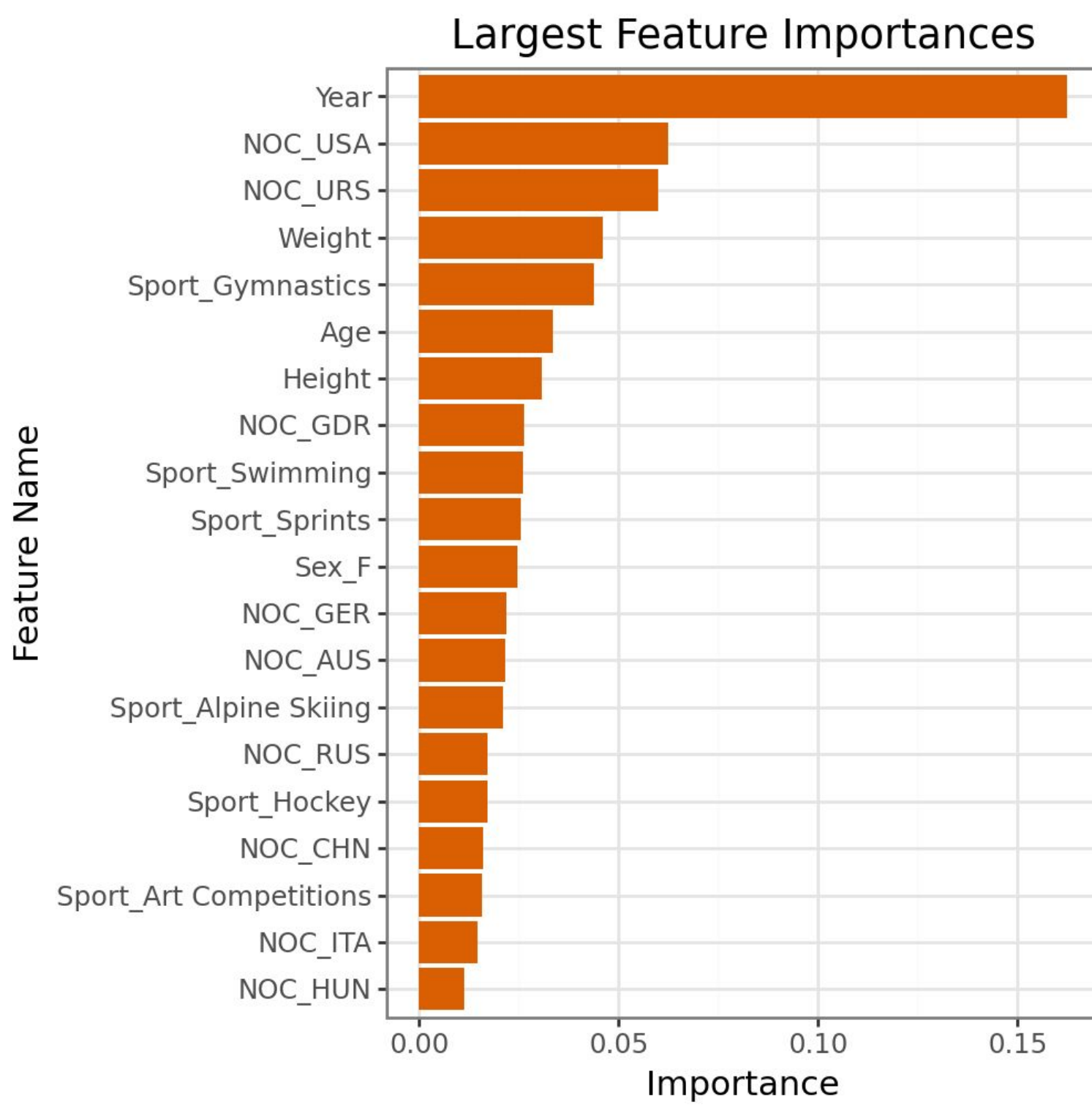


Fig 3: The top 20 highest decision tree feature importances of our global model. Feature importance measures how much an input feature contributes to the model’s predictions.

Models

Global Model: We chose the model with the best overall Mean CV F1 score during our grid search. This turned out to be a decision tree model with these hyperparameters:

- **Max depth** = 20
- **Minimum samples per leaf** = 5
- **Minimum samples per split** = 25

Overall, this model was able to achieve these evaluation scores on our data:

- **Mean CV F1:** 0.4342
- **Train F1:** 0.4687
- **Test F1:** 0.4418

Sport-Specific Models: We also tested out fitting models using data only from that sport in the dataset. Our idea was that sports have wildly different keys to success, whether it be athlete or other characteristics, and thus fitting per sport allows us to capture those better than generalizing over all the sports. We kept the same model type and hyperparameters as the global model in order to make a direct performance comparison on the data.

Results / Implications

Primary Question: Can we accurately predict if an athlete will win an Olympic medal despite extreme class imbalance?

To some extent, yes. Using the full dataset, the model achieved an F1 score of **0.45** when Year was included as a feature, 0.41 when not. While this introduced data leakage, we were fine with that, as our model wasn’t meant to generalize to future events. Precision was relatively low (0.34) compared to recall (0.60), meaning the model identified many medalists but produced many false positives. In other words, roughly **70%** of predicted medalists did not actually win a medal.

Performance improved when models were trained for individual sports. For example, the **rowing** model achieved an F1 score of **0.62** with moderate precision and high recall. Feature importance analysis also revealed meaningful differences between the global and sport-specific models. In rowing, features such as **Year (0.3)**, **Age**, and **Weight** were more influential, and countries like **Germany** appeared prominently, reflecting their historical strength in the sport. While these models may not generalize well to future Olympic events, they provide insight into country dominance, sport specialization, and the physical characteristics associated with success in specific sports.

Secondary Question: Are Olympic medal predictions more driven by individual athlete characteristics versus other factors?

We compared models using only **individual athlete characteristics** (Age, Height, Weight, Sex) with models using **other factors** (NOC, Season, Sport, Year). The athlete-only models performed much worse, with Test F1 scores of **0.2655** for Logistic Regression and **0.2835** for Decision Tree, while the other factors models reached **0.3890** and **0.4480**. This shows that medal predictions in our dataset are **more** strongly driven by contextual factors than by the athlete-level characteristics we included. That fits our data context, since Olympic outcomes reflect nationality, sport structure, historical era, and institutional conditions, not just athlete traits. At the same time, this should be interpreted carefully because **Year** is part of the other factors model, and our main project already noted possible leakage from that variable. The results suggest that Olympic success is better predicted by **broader structural context** than by the limited set of individual characteristics available in the data.

Ethics & Limitations

These models don’t measure pure athletic ability. Olympic medal outcomes also reflect unequal access to training, funding, coaching, and competition opportunities based on gender, class, race, nationality, and time. For instance, women were excluded from the 1896 Olympics and had limited events when first included in 1900, so historical medal patterns reflect exclusion rather than performance. The model may thus reproduce past inequalities rather than predict success. The data spans 1896-2016, combining different Olympic eras shaped by wars, boycotts, exclusions, and rule changes. For example, World War II canceled the Games, and postwar exclusions changed who could compete. Our project also notes likely data leakage from year, so some predictive power may come from learning historical eras instead of true athlete medal likelihood. Additionally, making medal outcome binary and removing detailed event information reduces the model’s nuance. The data was scraped from archived Olympic results, so the model can only be inferred to athlete event observations similar to those in the included dataset, not to all athletes or future competitions. It shouldn’t be used to judge an athlete’s worth or potential, as Olympic success depends on structural factors beyond the data’s capture. For instance, countries with stronger investments in certain sports may appear more predictable due to resources and support, not inherent athletic ability.